

7.3 General Concepts of Point Estimation

When measurements come from a probability distribution with unknown parameter, we estimate the unknown parameter using the measurement data.

Example 4

The numbers of 10 crashed aircraft are taken from an aircraft fleet of unknown size:

287 182 213 317 077 153 323 131 437 280

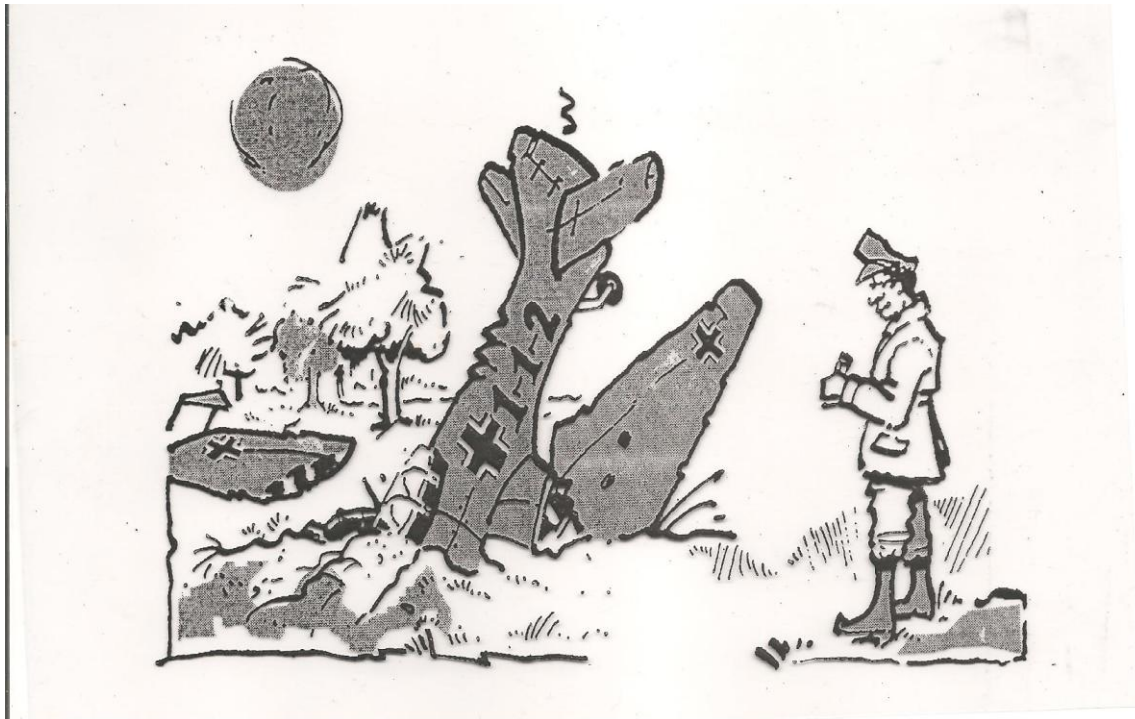


Figure 7.5

Collecting data

Give an estimate for the size of the aircraft fleet.

Solution

Suppose the aircraft numbers are a random sample of a uniformly distributed random variable X on the integers $a = 1 \leq x \leq b = \text{unknown}$.

Since

$$E(\bar{X}) = \frac{1}{2}(a + b) = \frac{1}{2}(b + 1),$$

$E(2\bar{X} - 1)$ must be equal to b . We therefore first determine the sample mean.

Because $\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{2400}{10} = 240$, $b \approx 2 \cdot 240 - 1 = 479$. So 479 is an estimate for b .

Definition

Let X be a random variable with **unknown parameter θ** . Let $X_1, X_2, \dots, X_n$ be a **random sample** of X .

1. The **statistic** $\hat{\theta} = g(X_1, X_2, \dots, X_n)$ is called an **unbiased estimator for θ** , if $E(\hat{\theta}) = \theta$.
In that case, the single numerical value $\hat{\theta} = g(x_1, x_2, \dots, x_n)$ is called an **unbiased estimate**.
2. If the parameter θ has several unbiased estimators, the estimator with the **smallest variance** is preferred. This estimator is called the **minimum variance unbiased estimator (MVUE)**.

Examples

5. Let $X_1, X_2, \dots, X_n$ be a random sample of an arbitrary random variable X with unknown parameter **$\mu = E(X)$** .

Since $E(\bar{X}) = \mu$, regardless the probability distribution of X ,

$$\hat{\theta} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{sample mean}$$

is an **unbiased estimator for μ** .

In particular:

- If $X \sim N(\mu = ?; \sigma^2 = \sigma_0^2)$ and $\bar{x} = 18.34$, then $\hat{\mu} = 18.34$. So $\mu \approx 18.34$.
 - If $X \sim u(a = 0; b = ?)$ and $\bar{x} = 5.2$, then $\hat{b} = 2 \cdot \bar{x} = 10.4$. So $b \approx 10.4$.
6. Let $X_1, X_2, \dots, X_n$ be a random sample of the random variable X , with **$X \sim \text{alt}(p)$** and **p** unknown.

Then

$$\hat{\theta} = \hat{P} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{K}{n} \quad \text{sample proportion}$$

is an **unbiased estimator for p** .

Proof

$$\begin{aligned} E(\hat{P}) &= E\left(\frac{K}{n}\right) \\ &= \frac{1}{n} \cdot E(K) \\ &= \frac{1}{n} \cdot np \\ &= p \end{aligned} \quad K \sim \text{bin}(n; p)$$

In particular, if K = the number of smokers $\sim \text{bin}(n; p)$, with $n = 50$ people and $k = 18$ smokers, then $\hat{p} = 18/50 = 0.36$. So $p \approx 0.36$.

7. Let $X_1, X_2, \dots, X_n$ be a random sample of an arbitrary random variable X with unknown parameter $\sigma^2 = V(X)$.

Then

$$\hat{\Theta} = S^2 = \frac{1}{n-1} \{ \sum_{i=1}^n X_i^2 - n \cdot \bar{X}^2 \} \quad \text{sample variance}$$

is an unbiased estimator for σ^2 .

Proof

$$\begin{aligned} E(S^2) &= E \left(\frac{1}{n-1} \{ \sum_{i=1}^n X_i^2 - n \cdot \bar{X}^2 \} \right) \\ &= \frac{1}{n-1} \{ \sum_{i=1}^n E(X_i^2) - n \cdot E(\bar{X}^2) \} \end{aligned}$$

Since $V(X) = E(X^2) - E(X)^2$, $E(X^2) = \sigma^2 + \mu^2$.

Thus

$$\begin{aligned} E(S^2) &= \frac{1}{n-1} \{ \sum_{i=1}^n (\sigma^2 + \mu^2) - n \cdot \left(\frac{\sigma^2}{n} + \mu^2 \right) \} \\ &= \frac{1}{n-1} \{ n \cdot (\sigma^2 + \mu^2) - \sigma^2 - n \cdot \mu^2 \} \\ &= \sigma^2 \end{aligned}$$

In particular, if $X \sim N(\mu = ?; \sigma^2 = ?)$, and $x_1 = 5.2, x_2 = 4.8, x_3 = 5.9$, then:

$$\begin{aligned} L_1 &= \{ 5.2, 4.8, 5.9 \} & \text{gc/stat/edit/1} \\ \bar{x} &= 5.3 & \text{gc/stat/calc/1/L}_1 \\ s &= 0.5568 & \text{gc/stat/calc/1/L}_1 \end{aligned}$$

So

$$\begin{aligned} \hat{\mu} &= 5.3 & (\text{i.e. } \mu \approx 5.3) \\ \hat{\sigma} &= 0.5568 & (\text{i.e. } \sigma \approx 0.5568) \end{aligned}$$

Remarks

1. If $X \sim N(\mu; \sigma^2)$, it can be proved that $\hat{\Theta} = \bar{X}$ is a **minimum variance unbiased estimator**.
2. The estimator $\hat{\Theta} = S^2$ is an unbiased estimator for the parameter σ^2 . It can be proved that S is a **biased estimator** for σ . The **bias** (i.e. $E(\hat{\Theta}) - \theta$) becomes smaller as n becomes larger.
3. The **precision** of the unbiased estimator $\hat{\Theta}$ is recorded using the **standard error (SE)**:

$$\sigma_{\hat{\Theta}} = SE(\hat{\Theta}) = \sqrt{V(\hat{\Theta})}.$$

The more precise the estimator, the smaller the standard error.

The standard error of $\bar{X}$ equals

$$\sigma_{\bar{X}} = \sqrt{V(\bar{X})} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}.$$

So the unbiased estimator $\bar{X}$ becomes more precise as the sample size n becomes larger.

4. An **unbiased** estimator is not always preferred to a **biased** estimator. This choice also depends on the precision.
5. In the event of a **biased estimator**, the precision is recorded using the **mean square error (MSE)**:

$$MSE(\hat{\Theta}) = E[(\hat{\Theta} - \theta)^2].$$

It is not difficult to reduce the MSE to:

$$MSE(\hat{\Theta}) = V(\hat{\Theta}) + \underbrace{(E(\hat{\Theta}) - \theta)^2}_{\text{bias}}.$$

6. So, in the case of an **unbiased estimator** $\hat{\Theta}$:

$$SE(\hat{\Theta}) = \sqrt{MSE(\hat{\Theta})}.$$

7. The **relative efficiency** of $\hat{\Theta}_2$ to $\hat{\Theta}_1$ is defined by

$$\frac{MSE(\hat{\Theta}_1)}{MSE(\hat{\Theta}_2)}.$$

If this relative efficiency is less than 1, $\hat{\Theta}_1$ is a more efficient estimator of θ than $\hat{\Theta}_2$.

Example 8

Let X be a random variable with the following probability distribution

$$f(x) = \frac{2x}{a^2}, \text{ for } 0 \leq x \leq a.$$

Find an **unbiased estimation** for a , based on a random sample of size 25, if $\bar{x} = 6.4$.

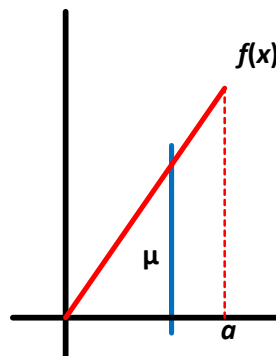


Figure 7.6

Probability density function for Example 8

Solution

$$\begin{aligned} E(\bar{X}) &= \mu = \int_0^a x \cdot \frac{2x}{a^2} dx \\ &= \left[\frac{1}{a^2} \cdot \frac{2}{3} x^3 \right]_0^a \\ &= \frac{2}{3} a \end{aligned}$$

So

$$E\left(\frac{3}{2}\bar{X}\right) = a$$

Thus

$$\begin{aligned} \hat{a} &= \hat{\theta} = \frac{3}{2} \cdot 6.4 \\ &= 9.6 \end{aligned}$$

So

$$a \approx 9.6$$